

City Pollution Reduction Planner - User Manual

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2023-08459
AB-3L

Introduction

What is the City Pollution Reduction Planner?

The City Pollution Reduction Planner is an interactive Shiny application (R-based data application) designed to help Greenvale's environmental commission optimize pollution reduction strategies. The application runs through R/RStudio and provides a web-like interface. It allows users to select from 30 available mitigation projects and evaluate their combined impact on reducing ten priority pollutants while minimizing total implementation costs.

The Problem

Greenvale faces a mandate from the national government to drastically reduce pollution within next year. The city must meet specific reduction targets for ten pollutants:

- **CO2** - tons
- **NO** - tons
- **SO2** - tons
- **PM2.5** - tons
- **CH4** - tons
- **VOC** - tons
- **CO** - tons
- **NH3** - tons
- **BC** - tons
- **N2O** - tons

With limited resources, the city must choose the most cost-effective combination of mitigation projects to meet all pollution reduction targets.

How the App Solves It

The application uses linear programming optimization (Simplex Method with Two-Phase approach) to find the minimum-cost solution that meets or exceeds all pollution reduction requirements. Each mitigation project can be deployed in units (up to 20 units per project, representing deployment across Greenvale's 20 barangays), and the optimizer determines the ideal number of units for each selected project.

Prerequisites

System Requirements

To use the City Pollution Reduction Planner, you need:

- **Web Browser:** Modern web browser (Chrome, Firefox, Safari, or Edge)
- **R Environment** (for running the application locally):
 - R version 4.0 or higher
 - RStudio (recommended)

Required R Packages

The application requires the following R packages:

- shiny, bslib, DT

Installation

If running locally, install the required packages:

r

None

```
install.packages(c("shiny", "bslib", "DT"))
```

Running the Application

1. Open RStudio (or R)
2. Set your working directory to the folder containing `app.R`:

r

None

```
setwd("path/to/your/app/folder")
```

3. Run the application using one of these methods:
 - **In RStudio:** Open `app.R` and click the "Run App" button at the top
 - **In R Console:** Type `shiny::runApp()`
4. The application will launch in your default web browser or RStudio's Viewer pane

How to Use

Getting Started

Step 1: Launch the Application

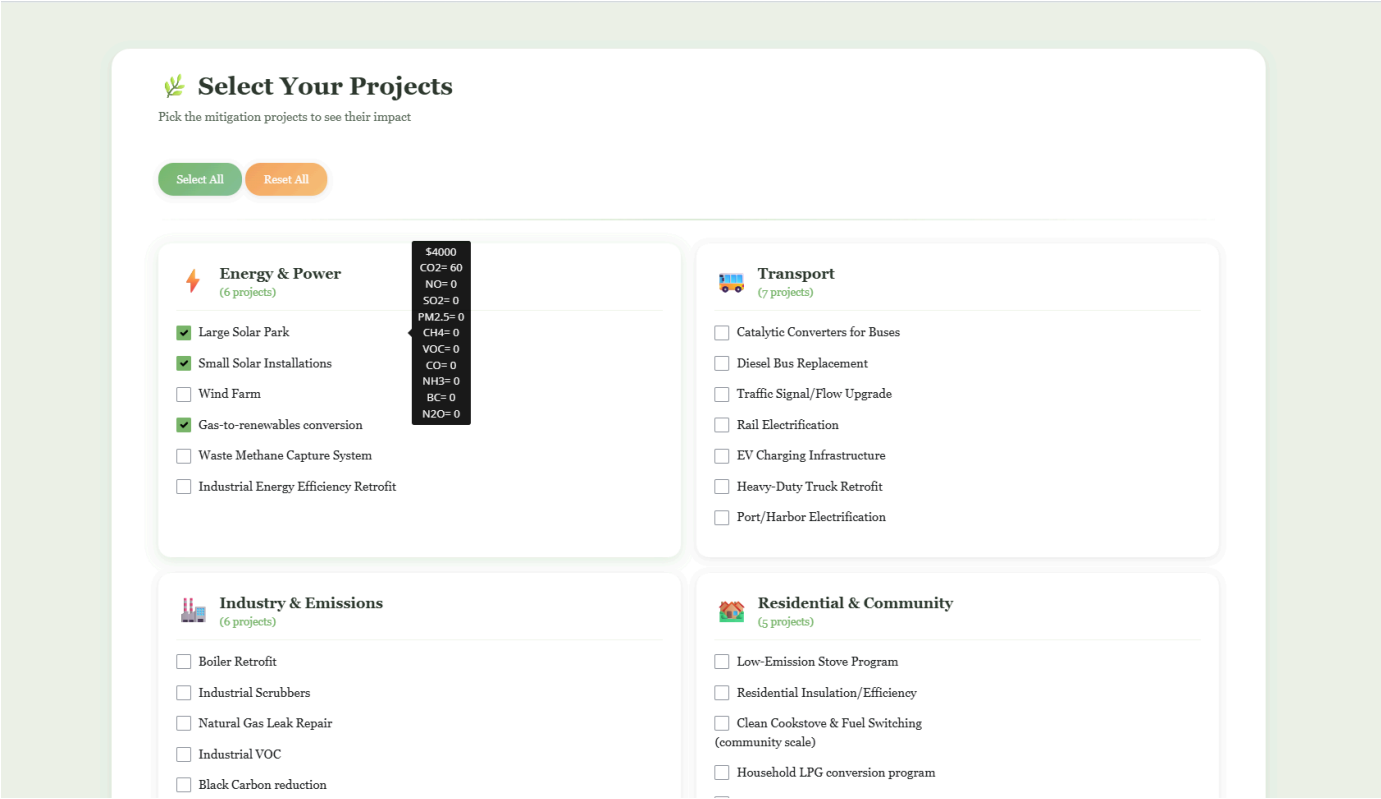
When you open the application, you'll see the **Welcome Screen** featuring:

- The application title: "City Pollution Reduction Planner"
- A description of Greenvale's pollution challenge
- A **"Get Started"** button



Click **"Get Started"** to proceed to the project selection interface.

Step 2: Project Selection



You'll now see the **"Select Your Projects"** page, which displays:

- **Select All** button (green) - selects all 30 projects
- **Reset All** button (orange) - deselects all projects
- **Five project categories**, each containing related mitigation options:
 - ⚡ **Energy & Power** (6 projects)
 - 🚗 **Transport** (7 projects)
 - 🏭 **Industry & Emissions** (6 projects)
 - 🏠 **Residential & Community** (5 projects)
 - 🌱 **Land Use & Agriculture** (6 projects)

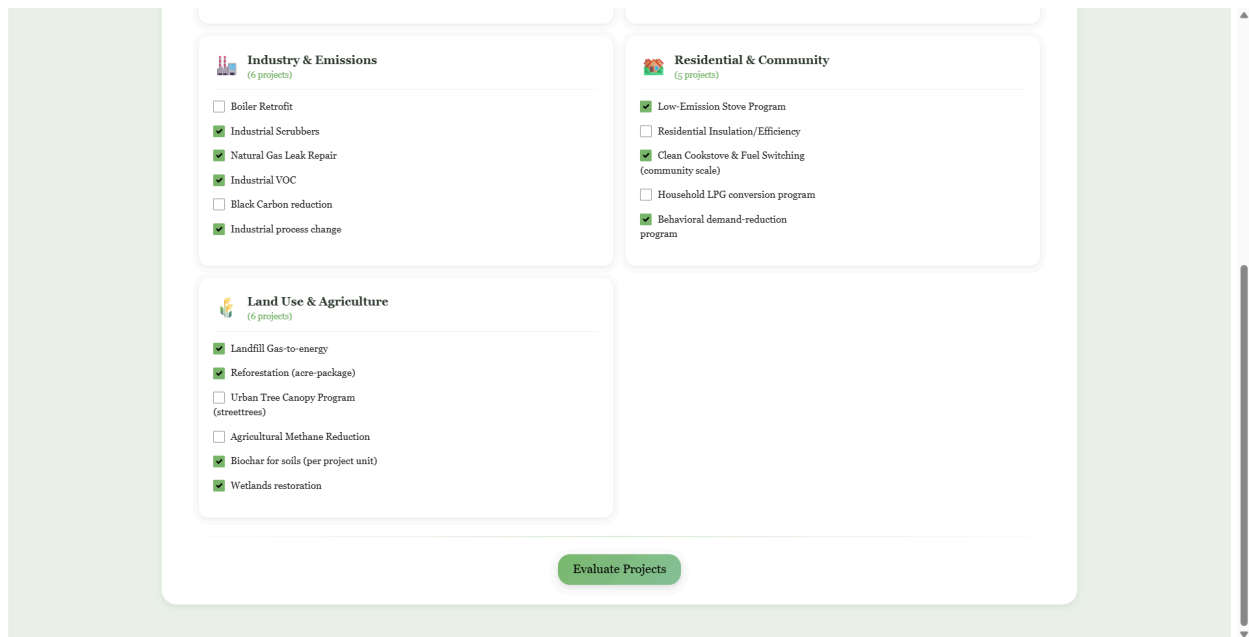
Each category card shows the number of projects it contains and lists individual mitigation options with checkboxes.

Selecting Projects

1. **Browse through each category** to review available projects
2. **Hover over any project name** to see a tooltip displaying:
 - Project cost per unit
 - Pollution reduction values for each of the 10 pollutants
3. **Click the checkbox** next to any project you want to include
4. **Check multiple projects** across different categories as needed

Evaluating Your Selection

Step 3: Run the Evaluation



Once you've selected your desired projects:

1. Scroll to the bottom of the project selection page
2. Click the large green **"Evaluate Projects"** button
3. The application will process your selection and display results

Step 4: Review the Results

After evaluation completes, you'll be automatically taken to the "Selected Projects Overview" page. This page contains all your results organized in several sections:

← Back to Planner

Selected Projects Overview

Here's how your selected projects reduce pollution

You selected 24 mitigation projects

• Large Solar Park

• Small Solar Installations

• Wind Farm

• Gas-to-renewables conversion

• Waste Methane Capture System

• Industrial Energy Efficiency Retrofit

• Catalytic Converters for Buses

• Diesel Bus Replacement

• Traffic Signal/Flow Upgrade

• Rail Electrification

• EV Charging Infrastructure

• Heavy-Duty Truck Retrofit

• Port/Harbor Electrification

• Industrial Scrubbers

• Natural Gas Leak Repair

• Industrial VOC

• Industrial process change

• Low-Emission Stove Program

• Clean Cookstove & Fuel Switching (community scale)

• Behavioral demand-reduction program

• Landfill Gas-to-energy

• Reforestation (acre-package)

• Biochar for soils (per project unit)

• Wetlands restoration

Optimized Cost: \$605,000

Search:

Solution and Cost Breakdown

Mitigation Project	Number of Project Units	Cost (\$)
Gas-to-renewables conversion	20	64000
Catalytic Converters for Buses	20	52000
Diesel Bus Replacement	20	18+05
Traffic Signal/Flow Upgrade	20	20000
Industrial VOC	5+5	13383+18
Wetlands restoration	20	36000
Household LPG conversion program	20	14000
Industrial process change	4+34	21704+09
Behavioral demand-reduction program	20	8000

Search:

Pollution Reduction Summary

Pollutant	Total Reduction	Target Reduction
CO2	1391+72	1000
NO	35	35
SO2	25	25
PM2.5	49+86	20
CH4	267+86	60
VOC	45	45
CO	194+41	80
NH3	12	12
BC	12+06	6
N2O	10	10

Show Iteration Tablans

At the top:

- "← Back to Planner" button (orange) - returns you to project selection with your choices preserved

Main content sections:

1. **Your Selection Summary** - Shows:
 - Number of projects you selected (e.g., "You selected 24 mitigation projects")
 - List of all selected project names
2. **Optimized Cost Display** - Shows one of two outcomes:
 - **Feasible**: "Optimized Cost: \$605,000" (in green)
 - **Infeasible**: "This selection is infeasible. The target pollution reductions cannot be met with the chosen projects." (in red)
3. **Solution and Cost Breakdown** table - Displays:
 - Which projects are used in the optimal solution

- How many units of each project (0-20)
 - Individual and total costs
 - Searchable to quickly find mitigation project
4. **Pollution Reduction Summary** table - Displays:
- All 10 pollutants (CO₂, NO, SO₂, PM_{2.5}, CH₄, VOC, CO, NH₃, BC, N₂O)
 - Total reduction achieved and target reduction required
 - For infeasible solutions: shows "Maximum Possible Reduction" instead
 - Searchable to quickly find specific pollutants
5. **"Show Iteration Tableaus"** button at the bottom - Opens a modal showing the step-by-step Simplex algorithm calculations

EV Charging Infrastructure

90

Approx

Biochar for soils (per project unit)

20

offsets

Heavy Duty

Woolfend

Behavior

Full

CO₂

NO

SO₂

PM_{2.5}

CH₄

VOC

CO

NH₃

BC

N₂O

Iteration Tableaus

Phase 1 - Iteration 1

Iteration-	Cj	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
1																				
B	CB	XB	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10	x11	x12	x13	x14	x15	x16	x17	x18
s1	0	20	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s2	0	20	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s3	0	20	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s4	0	20	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
s5	0	20	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
s6	0	20	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
s7	0	20	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
s8	0	20	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
s9	0	20	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
s10	0	20	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
s11	0	20	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
s12	0	20	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
s13	0	20	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
s14	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
s15	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
s16	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Functionalities

1. Results Overview

After evaluation, you'll see the **"Selected Projects Overview"** page with several key sections:

A. Selected Projects List

- **Total count** of selected projects (e.g., "You selected 24 mitigation projects")
- **Two-column layout** displaying all selected project names
- Easy reference to confirm your selections

B. Optimized Cost Display

Two possible outcomes:

Feasible Solution:

Optimized Cost: \$605,000

- Displayed in green
- Shows the minimum cost to meet all pollution targets with your selected projects

Infeasible Solution:

This selection is infeasible. The target pollution reductions cannot be met with the chosen projects.

- Displayed in red
- Indicates your project selection cannot meet all required reduction targets
- You'll need to select additional or different projects

2. Solution and Cost Breakdown Table

Shows exactly how many units of each project should be implemented and at what cost.

Table Columns:

- **Mitigation Project:** Name of the project
- **Number of Project Units:** Optimal number of units to deploy (0-20)
- **Cost (\$):** Total cost for that project's deployment

Features:

- Searchable (use the search box in the top right)
- Shows only projects that are part of the optimal solution
- Clear breakdown of where budget will be allocated

3. Pollution Reduction Summary Table

Compares achieved pollution reductions against required targets for all 10 pollutants.

Table Columns:

For Feasible Solutions:

- **Pollutant:** Name of the pollutant (CO2, NO, SO2, etc.)
- **Total Reduction:** Actual reduction achieved by the optimal solution
- **Target Reduction:** Required reduction to meet government mandate

For Infeasible Solutions:

- **Pollutant:** Name of the pollutant
- **Maximum Possible Reduction:** Best possible reduction with selected projects
- **Target Reduction:** Required reduction (showing the gap)

Features:

- Searchable
- Allows you to verify that all targets are met
- For infeasible solutions, identifies which pollutants fall short

4. Iteration Tableaus

To understand the optimization process, this feature displays the step-by-step iterations of the Two-Phase Simplex Method algorithm.

Phase 1 Iterations:

- Initial tableau setup with artificial variables
- Each iteration shows the tableau matrix

- Process notes (e.g., "entering variable," "leaving variable")
- Continues until all artificial variables are eliminated
- Concludes with: "Since all $C_j - Z_j \geq 0$, phase 1 is done"

Phase 2 Iterations:

- Begins with the feasible solution from Phase 1 and the original objective function
- Shows optimization toward minimum cost
- Each iteration moves closer to the optimal solution
- Final tableau represents the optimal solution

For Infeasible Solutions:

- Phase 1 iterations only
- Message: "This solution is not feasible because an artificial variable appears in the basis with positive value"

5. Navigation Features

Back to Planner Button

- Located at the top of the results page (orange button with arrow)
- Returns you to the project selection screen
- **Preserves your previous selections** - your checkboxes remain checked
- Allows you to modify your project selection and re-evaluate

Select All / Reset All

Available on the project selection screen:

- **Select All:** Instantly check all 30 projects (useful for finding the absolute best possible solution)
- **Reset All:** Clear all selections to start fresh

6. Tooltips and Interactive Elements

Project Tooltips - Hover your mouse over any project name in the project selection

Information displayed:

- **Cost per unit** (e.g., "\$4000")
- **Pollution reductions** for all 10 pollutants (e.g., "CO₂= 60, NO= 0, SO₂= 0...")

Interactive Tables

- **Search functionality:** Filter rows by typing keywords
- **Responsive design:** Tables adjust to your screen size

References:

For two-phase simplex algorithm:

<https://cbom.atozmath.com/example/CBOM/Simplex.aspx?q=tp&q1=E1>

For background image:

<https://www.jinding.com.au/news/jinding-plans-425-lot-greenvale-community-following-highly-successful-year/>